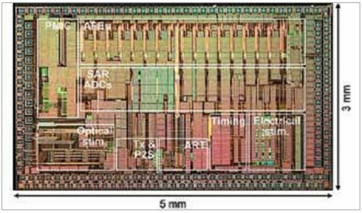
thin enough to preserve the original size, form, and structure of the objects. After coating the object with M-spray, it was magnetised in single or multiple magnetisation directions, which could control how the object moved in a magnetic field. Then heat was applied to the object until the coating solidified.

As the magnetic coating is biocompatible and can be disintegrated into powder, when needed, this technology demonstrates the potential for biomedical applications, including catheter navigation and drug delivery.

## For neural research, wireless chip shines light on the brain




*wireless chip (Credit: https://news.ncsu.edu)*

Researchers have developed a chip that is powered wirelessly and can be surgically implanted to read neural signals and stimulate the brain with both light and electrical current. The technology has been demonstrated successfully in rats and is designed for use as a research tool for understanding the behaviour of different regions of the brain, particularly in response to various forms of neural stimulation, which can also help pave the way for advances in addressing neurological disorders such as Alzheimer's or Parkinson's disease.

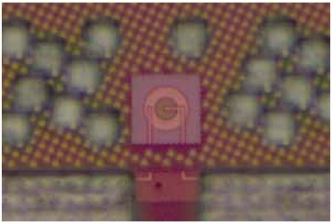
Besides being capable of sending and receiving information wirelessly, the chip can read neural signals in targeted regions of the brain by detecting electrical changes in those regions and stimulate the brain by introducing a small electrical current into the brain tissue.

## LED that can be integrated directly into computer chips

LEDs are not just simple inexpensive light sources but are useful for microelectronics too. However, it is tough to make LEDs from silicon. That means LED sensors must be manufactured separately from their device's silicon-based processing chip, often at a hefty price.

But that could one day change thanks to new research from MIT's Research Laboratory of Electronics (RLE). The MIT researchers have fabricated a silicon chip with fully integrated LEDs, bright enough to enable state-of-the-art sensor and communication technologies. The advance could lead to not only streamlined manufacturing but also better performance for nanoscale electronics.

The silicon-based LED has been designed with specially engineered junctions that enhance brightness and efficiency: The LED operates at low voltage, but it still produces enough light to transmit a signal through 5 metres of fibre-optic cable. The ability of the LED to send signals at frequencies up to 250MHz indicates that the technology could potentially be used not only for sensing applications but also for efficient data transmission. Thus, silicon integrated circuits will be able to communicate with one another directly with light instead of electric wires.



*MIT researchers have developed a bright, efficient silicon LED, pictured, that can be integrated directly onto computer chips. The advance could reduce cost and improve performance of microelectronics that use LEDs for sensing or communication (Credit: https://news.mit.edu)*

## Flexible rechargeable battery that's ten times more powerful

A team of researchers from the University of California in San Diego and California-based company ZPower has developed a flexible, rechargeable silver-oxide-zinc battery with five to ten times greater areal energy density of 50 milliamps per square centimetre at room temperature than state of the art. The battery is also easier to manufacture; while most flexible batteries need to be manufactured in sterile conditions, under vacuum, this one can be screen-printed in normal lab conditions.

The battery can be used in flexible, stretchable elec-